

AI-Powered Lecture Shortener Framework

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Abstract. The exponential growth of educational video content presents challenges for efficient knowledge consumption. This paper presents an AI-powered lecture shortener framework that automatically generates condensed versions of educational videos by processing both audio-visual content and accompanying presentation slides. Our system employs a three-stage pipeline: (1) video processing with speech-to-text conversion using Whisper, (2) slide processing with OCR and text summarization using Gemini, and (3) final video assembly with speech-slide matching and text-to-speech synthesis. The framework addresses the critical need for time-efficient learning by extracting key information from lengthy lectures while maintaining educational coherence. Experimental results demonstrate the system’s capability to produce meaningful condensed content that preserves essential information while significantly reducing viewing time. This work contributes to the emerging field of automated educational content curation.

1. Introduction

The digital transformation of education has produced an overwhelming abundance of video content from online courses, lecture recordings, and educational channels, creating challenges for learners who must balance time constraints with knowledge acquisition. Medical students preparing for exams may face 20 hours of lectures they cannot realistically review in full, while working professionals pursuing certificates often struggle to fit comprehensive materials into limited evening study sessions. In corporate training, mandatory compliance modules often extend to hours despite containing only brief segments of essential information, resulting in productivity losses when applied across large workforces. Distance learners, particularly those in different time zones, must contend with lengthy recorded seminars that exceed their available study time, limiting their ability to engage fully with course content. Similarly, researchers seeking to review dozens of conference presentations face weeks of viewing that could be reduced dramatically if condensed versions highlighting key contributions were available. Even educators encounter inefficiencies, as producing both full-length and shortened versions of lectures requires significant manual effort and often yields inconsistent quality. Across these contexts, the ability to automatically generate focused, high-quality summaries that preserve critical concepts would substantially improve accessibility, efficiency, and learning outcomes for students, professionals, researchers, and content creators alike.

Traditional approaches to content summarization have primarily targeted text-based materials. Early extractive techniques such as TextRank [Mihalcea and Tarau \(2004\)](#) and LexRank [Erkan and Radev \(2004\)](#) rely on graph-based ranking to identify salient sentences, while neural abstractive approaches like BART [Lewis et al. \(2020\)](#) employ pretrained sequence-to-sequence architectures to generate more fluent summaries. More recently, large language models such as GPT-4 [OpenAI \(2023\)](#) have redefined summarization as a general instruction-following task, achieving strong performance across diverse domains. However, these methods are largely optimized for text and fall short when applied to multimodal educational content, where spoken language, visual slides, and temporal dynamics must be preserved to maintain pedagogical effectiveness.

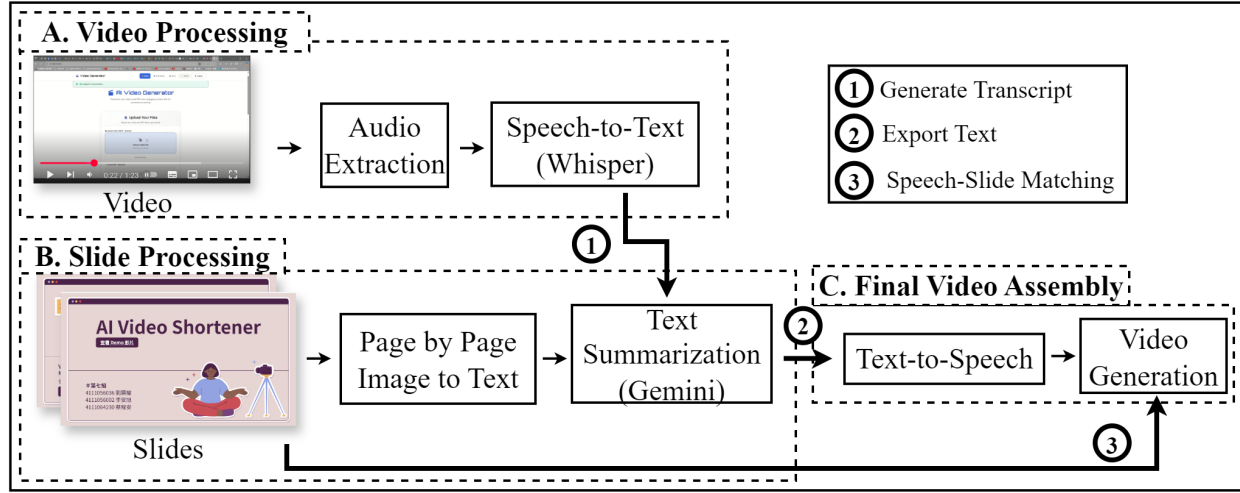


Figure 1. Complete workflow of the AI-Powered Lecture Shortener Framework, showing the three main processing stages: (A) Video Processing, (B) Slide Processing, and (C) Final Video Assembly.

This paper introduces an AI-powered lecture shortener framework that tackles the challenges of time-constrained learning through intelligent multi-modal processing. By automatically analyzing educational videos alongside their corresponding presentation slides, the system generates condensed versions that preserve essential information while significantly reducing consumption time. Key contributions include a novel framework that jointly processes video and slide content, an intelligent speech-slide matching algorithm to ensure coherence, a practical solution tailored to diverse learning scenarios, and a fully automated approach that removes the need for manual content condensation. Together, these innovations mark a substantial advancement in educational content curation, offering a scalable and efficient tool for learners, educators, and professionals seeking streamlined access to knowledge.

2. Methods

Our AI-powered lecture shortener framework operates through a three-stage pipeline designed to process educational videos comprehensively while maintaining semantic coherence between audio and visual components. The architecture is illustrated in Figure 1.

2.1. Video Processing

The video processing stage extracts and processes audio content from input lecture videos. This stage consists of two primary components: audio extraction and speech-to-text conversion.

2.1.1. AUDIO EXTRACTION

The system first extracts audio tracks from input videos using FFmpeg, a robust multimedia processing framework. This step handles various video formats and ensures consistent audio quality for subsequent processing. The audio

extraction process normalizes sampling rates and applies noise reduction techniques to optimize performance of downstream speech recognition models.

2.1.2. SPEECH-TO-TEXT CONVERSION

We employ OpenAI’s Whisper model(Radford et al., 2022) for automatic speech recognition due to its exceptional performance on diverse audio conditions and multilingual capabilities. Whisper’s transformer-based architecture provides robust transcription quality even in the presence of background noise, varying speaking speeds, and technical terminology common in educational content.

The speech recognition process generates time-stamped transcripts that preserve temporal alignment with the original video. These timestamps prove crucial for the subsequent speech-slide matching process, enabling precise synchronization between audio content and visual presentations.

2.2. Slide Processing

The slide processing stage analyzes presentation slides to extract textual content and generate summaries that can be integrated with the audio transcription. This stage employs direct PDF text extraction followed by intelligent text summarization.

2.2.1. PAGE-BY-PAGE IMAGE TO TEXT

Each slide is processed individually using Poppler, a PDF rendering library, to extract textual content directly from PDF presentations. The system handles various slide formats and layouts commonly found in educational presentations while maintaining the original text structure and formatting.

Poppler generates structured text output that preserves the hierarchical organization of slide content, including headings, bullet points, and other formatting elements that contribute to semantic understanding.

2.2.2. TEXT SUMMARIZATION WITH GEMINI

The extracted text from all slides undergoes intelligent summarization using Google’s Gemini model (Team, 2023). This large language model demonstrates exceptional capability in understanding technical content and generating coherent summaries that preserve key concepts while reducing verbosity.

The summarization process considers the educational context of the content, prioritizing concept definitions, key principles, and practical examples while reducing redundant explanations and transitional content. The model’s contextual understanding enables it to maintain logical flow and conceptual coherence across the summarized content.

2.3. Final Video Assembly

The final stage integrates processed audio and slide content to generate the condensed video output. This stage implements sophisticated speech-slide matching algorithms and employs text-to-speech synthesis for seamless content integration.

2.3.1. SPEECH-SLIDE MATCHING

A critical component of our framework is the intelligent alignment of audio transcriptions with slide content. This process identifies corresponding segments between spoken content and visual presentations, enabling the system to maintain pedagogical coherence in the final output.

The matching algorithm employs semantic similarity measures to identify relationships between transcribed speech segments and slide text content. We utilize embedding-based approaches that capture semantic relationships beyond simple keyword matching, enabling robust alignment even when speakers use different terminology than what appears on slides.

The temporal information from the speech recognition process provides additional constraints that improve matching accuracy. By considering the sequential nature of presentations, the algorithm can resolve ambiguities and ensure logical progression in the final condensed content.

2.3.2. TEXT-TO-SPEECH SYNTHESIS

The final condensed script, generated through the integration of summarized speech and slide content, undergoes text-to-speech synthesis to create the audio track for the shortened video. We employ advanced neural text-to-speech models that generate natural-sounding speech with appropriate intonation and pacing for educational content.

The synthesis process considers the technical nature of educational content, ensuring proper pronunciation of specialized terminology and maintaining appropriate speaking rhythm for comprehension. The generated audio maintains consistency with the educational context while providing clear and engaging delivery.

2.3.3. VIDEO GENERATION

The final video assembly combines the synthesized audio with temporally aligned slide content to create the condensed educational video. This process maintains visual coherence by ensuring that slides are displayed at appropriate times corresponding to the synthesized narration.

The system generates metadata that documents the compression ratio, key topics covered, and temporal mapping between original and condensed content. This information enables users to understand the summarization process and facilitates further customization based on specific learning objectives.

3. Conclusion

This paper presents an AI-powered lecture shortener that streamlines digital learning by combining speech recognition, text summarization, and content synthesis to produce condensed yet pedagogically effective videos. Through a three-stage pipeline of video processing, slide analysis, and final assembly, the framework processes both audio and visual components without manual intervention. Its modular design integrates Whisper for transcription, Gemini for summarization, and intelligent speech-slide alignment, showcasing the potential of coordinated AI systems for complex educational tasks. As digital content proliferates, such automated curation tools provide a practical foundation for enhancing learning efficiency and enabling effective knowledge transfer

in time-constrained environments.

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